# The Fine-Structure Constant as an OPH Pixel Fixed Point

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**Abstract**

This paper studies a conditional pixel fixed point for the fine-structure constant. The same-cell equation $P = \varphi + \sqrt{\pi}\,\alpha(0)$ relates a screen-area coordinate to an electromagnetic readout, and a declared source map returns the electromagnetic readout from the same coordinate through declared closure coordinates and running conventions. Two declared trial maps have interval-certified unique fixed points, with inverse couplings $136.994835177413\ldots$ and $137.035660136947\ldots$, about 300 and 2.5 parts per million from the CODATA 2022 value. The first transports the electroweak anchor to the Thomson endpoint with a declared quark-screening ansatz; the second adds the finite-screen unified gauge width inside the loop. Replacing the ansatz by the measured hadronic spectral payload gives 136.3827548 on the interval $[136.3670481, 136.3984652]$, which excludes the measured value; reaching it would need a hadronic shift of 0.032244 instead of the measured $0.027609 \pm 0.000112$, or a same-scheme source-anchor bridge of about 0.635 inverse-alpha units. The calculation is a conditional diagnostic, not a same-process derivation from the executable model: its inputs include a supplied charged-fermion spectrum, a declared gauge-running convention and measured hadronic spectral data.

An independent entangled-pair construction due to A. D. Wissner-Gross gives the exact scalar $(79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6})/31104$, whose inverse is $137.035999189566\ldots$. We reconstruct its finite exchange/projector arithmetic and complete measurement controls; if its conjectured Thomson identification and the same-cell reading are granted, it supplies another conditional area coordinate. The constructions supply no source-selected photon normalization, hadronic threshold transport, absolute particle masses or physical same-cell bridge, and every numerical endpoint comparison is a retrospective diagnostic.

**Keywords:** fine-structure constant, Observer-Patch Holography, pixel fixed point, electromagnetic coupling, Standard Model constants, hadronic vacuum polarization

**One theory and its executable hypothesis.** OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

*A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed $3+1$-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

**The tower.** The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored

extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

**Completion and stopping.** The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

**Fixed physical boundaries.** Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

## This paper's hypothesis and place in the tower

This paper carries a conditional branch at layer seven. Its hypothesis is a pixel fixed point: if a typed bridge proves that the detuning $(P-\varphi)/\sqrt{\pi}$ of one screen cell from the golden-ratio balance and the Thomson-limit coupling $1/A_{\mathrm{Th}}(P)$ returned by a declared source map at the same $P$ read one cell quantity, then closure requires $P = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$, and $\alpha(0)$ is the coupling at the fixed point. The source map consumes declared closure coordinates and conventions: the golden-ratio balance point, the Gaussian width $\sqrt{\pi}$, the $P/4$ representation-entropy budget, the one-loop MSSM coefficient triple, the transmutation coefficient $\beta_{\mathrm{EW}} = 4$, a supplied phenomenological charged-fermion spectrum and a declared quark-screening ansatz. The measured hadronic spectral payload enters only as a replacement test of that ansatz, and the CODATA value only as the endpoint.

The finite results are exact: each declared map has one interval-certified fixed point on its maximal analytic domain; coarse register data do not determine the hadronic residual, and a 24-site Jacobi realization of a chosen residual is tautological; the conditional pure-source theorem is Banach's

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/COMPLETION_PROGRAM.md

theorem under its listed premises. The calculation is not a same-process derivation from the executable model. That model supplies no physical area map for its screen cells, no Lorentzian Maxwell theory, no charged spectrum and no hadronic spectral measure, and it uses the source root only as the closure input of one detector.

The open joins are a target-independent rule selecting one map, the typed same-cell bridge, a native area and clock dictionary for $P$, charged masses from native defects, and a source-emitted Ward-current hadronic spectral measure with its same-scheme remainder. With the measured hadronic payload in place of the ansatz, the declared transport excludes the measured value, so the branch needs a same-scheme anchor bridge of about 0.635 inverse-alpha units. A target-independent selected map whose certified prediction interval, narrower than the direct rubidium measurement, excludes the measured value would refute the branch.

# 1 Introduction

The fine-structure constant is usually treated as an empirical constant. Here it is studied as a conditional readout of one screen cell. On the declared carrier branch the cell belongs to the federated screen built from twelve-port icosahedral carriers, the structure whose port algebra witnesses the Standard Model gauge structure in the gauge paper [2]. The known physics is the low-energy Maxwell coupling, charged-lepton running and low-energy hadronic spectral transport.

Two declared principles carry the construction. The detuning law places the pixel ratio off the golden-ratio balance point by the Gaussian-normalized width of one electromagnetic observation, $P = \varphi + \sqrt{\pi}\,\alpha$. The constant-identification law is a closure premise: a typed bridge must prove that the fine-structure constant read inside the trial world and the substrate pixel readout are one quantity. Conditional on that bridge, the two principles leave one local variable and one self-read equation. Neither the fixed-point mathematics nor the source certificates construct the bridge or select between the two certified maps. The measured fine-structure constant enters only after the source calculation, as an endpoint test.

Jentschura and Nandori survey first-principles attempts to understand $\alpha$, including beta-function, symmetry, cutoff and string-inspired ideas [14]. Golden-ratio constructions appear in geometric proposals [15] and in semi-empirical work on electroweak and flavor mixing [17]. Octonionic and exceptional-algebra approaches also produce values near 1/137 [16], and symbolic regression and quantum-information criteria have been used to look for structure in Standard Model constants and electroweak parameters [18, 19]. The calculation here belongs to this search for structure. Its physical interpretation requires a target-independent choice of the map, a typed bridge proving that its input and output read the same local screen cell, and same-scheme transport to the laboratory endpoint.

**Certified source roots.** The executable source map contracts to the undressed source/root inverse coupling $\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots$, the interval-certified unique fixed point of the declared map (public interval certificate [4]; enclosure width $7.2\times10^{-24}$, $L \le 0.0724$), 300.388218 parts per million below the CODATA 2022 value [6]. Its hadronic transport is the declared quark-screening ansatz $S_{\mathrm{calc}} \approx 0.887$ of Step 14, which carries no perturbative derivation. The self-consistent map that adds the finite-screen unified gauge width $\alpha_U(P)$ inside the loop has the certified fixed point $137.035660136946577\ldots$, with relative residual $2.5 \times 10^{-6}$, about $1.6 \times 10^4$ measurement sigma. The combination $A^{\mathrm{fp}}_{\alpha_U} = 137.035959513609\ldots$ mixes the certified source root with $\alpha_U(P_{\mathrm{C}})$ evaluated at the CODATA-derived comparison pixel; it is not a fixed point

of a single declared map.

**CODATA/NIST measured endpoint.** CODATA/NIST reports the value 137.035999177(21) for $\alpha^{-1}(0)$. The difference from $A^{\mathrm{fp}}_{\alpha_U}$ is CODATA-minus-diagnostic bookkeeping; it is not the output of a same-scheme hadronic spectral calculation.

The proximity of both roots to the measured value depends on the screening ansatz: replacing it by the measured hadronic payload gives 136.3827548, whose certified interval excludes the measured value (Section 22). The residual is therefore not explained by missing hadronic transport, and the comparison does not identify the physical source of the gap.

**Selection exposure and numerical support.** Two declared maps have interval-certified fixed points. A separate asymptotic compressed calculation prints 136.9940207, but is an approximate candidate: its printed grain and inverse coupling disagree in the outer identity by about $5.25 \times 10^{-6}$ relatively. It is not a third certified root. A separate retrospective constant-substitution grid contains 48 labels, 42 distinct numerical pairs and 41 alternatives to the canonical pair; none of those alternatives reaches the declared relative distance $2.5 \times 10^{-6}$. That threshold was chosen after the canonical distance was known. These counts describe numerical coverage, not a probability of accidental agreement. No exhaustive map-form menu or sampling measure is specified. Even the same two certified outputs, with one inside that distance threshold, give hit probability 1/2 under equal weights and 3/4 under weights $(1/4, 3/4)$. Those are mathematical counterexamples to identifying significance from output values and counts; neither weighting is inferred for this calculation. The measured endpoint therefore assigns no calibrated chance probability to the choice of map.

**Provenance of the three numbers.** The chain

$$136.994835177413\ldots \longrightarrow 137.035660136946577\ldots \longrightarrow 137.035999177(21)$$

runs from the source pixel through the gauge-width dressed pixel to the measured low-energy value. The first two values are the interval-certified fixed points of the declared source map and of the declared self-consistent gauge-width map [4]; the third is the measured endpoint. The root value $\alpha^{-1}_{\mathrm{root}}$ is returned by the incomplete source map before the low-energy charged vacuum is attached. The gauge width $\alpha_U$ is the main high-scale gauge-sector width carried by the pixel: the weak and color representation content must fit the pixel budget $P/4$, and the electroweak heat-kernel closure supplies the executable certificate. It supplies almost all of the inverse-alpha gap from the naked source value.

The factor $C_{24,Q}$ of Step 18 is a back-solved calibration: the small endpoint dressing required after the gauge-width contribution is added, solved from the measured endpoint rather than computed from source data. Its excess $\alpha_U(P_{\mathrm{C}})(C_{24,Q} - 1)$ is a comparison residual with the numerical size required of the residual same-scheme low-energy hadronic endpoint transport; it does not compute that transport. A zero-momentum laboratory photon sees the Ward-projected $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and finite endpoint matching, so the observer measures the fully dressed low-energy coupling, while $A^{\mathrm{fp}}_{\alpha_U}$ is a mixed-provenance no-hadron diagnostic before that dressing.

The observer account is the flagship synthesis [1], the source map and fixed-point branch are recorded in the Standard Model gauge paper [2], the charged-spectrum and particle-structure con-

tinuation in the particle paper [3], and the positive-chamber balance identity used by the charged-lepton continuation in the Koide paper [5].

**Proposition 1** (Maxwell normalization of the endpoint branch)**.** *On the declared electromagnetic branch used by the fixed-point equation, the Ward-projected* $\mathrm{U}(1)_Q$ *channel carries*

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2(q^2;P){*}J_Q,$$

*with*

$$g_Q^2(q^2;P) = 4\pi\alpha_{\mathrm{em}}(q^2;P), \qquad A_{\mathrm{Th}}(P) = \alpha_{\mathrm{em}}^{-1}(0;P).$$

*Thus the inside reading in the pixel equation is the Thomson-limit Maxwell coupling of the same electromagnetic current.*

*Proof.* On the declared unbroken Maxwell action of the gauge paper, the electromagnetic branch is the abelian factor $\mathrm{U}(1)_Q$. There the compact-gauge curvature reduces to $F_Q = dA_Q$, and the quadratic field action gives $d{*}F_Q = g_Q^2{*}J_Q$. That action is a branch premise: the executable observer supplies finite Maxwell-shaped seam actions, and a local source-produced physical Maxwell theory on Lorentzian spacetime is not constructed [8]. The particle-paper transport theorem then identifies the same $g_Q(q^2;P)$ with the Ward-projected running electromagnetic coupling. Taking $q^2 \to 0$ gives the endpoint quantity $A_{\mathrm{Th}}(P)$ used in the pixel fixed-point equation. □

The hadronic step uses the standard dispersion logic behind hadronic vacuum-polarization work. The relevant comparison data are electromagnetic spectral functions measured through $e^+e^- \to$ hadrons, as used in data-driven running-$\alpha$ and $g-2$ analyses [20, 21, 24]. In this paper, the decimal correction shown above is reported as the difference between the CODATA/NIST comparison value and the OPH pure source calculation. The source data in this paper emit no direct production calculation of the hadronic spectral function; a pure source endpoint theorem consumes that calculation as a premise.

## 2 Symbols

| Symbol | Definition | Informal meaning |
|---|---|---|
| $\mathcal{A}(P)$ | Patch algebra assigned to a screen patch $P$ | The local observables available to one finite observer patch. |
| $\omega_P$ | State on $\mathcal{A}(P)$ | The local physical description carried by that patch. |
| $\mathcal{A}(P \cap Q)$ | Shared overlap algebra | The observables two neighboring patches can compare. |
| $P$ | $a_{\mathrm{cell}}/\ell_\star^2$ | The dimensionless area of one screen cell in scale-readout units. |
| $a_{\mathrm{cell}}$ | Declared screen-cell area | The area assigned to the local pixel by the declared screen dictionary. |
| $\ell_\star$ | OPH scale-certificate length | The length emitted by the selected no-$G$ scale certificate. |
| $\ell_P$ | Planck length | The SI display of $\ell_\star$ after $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$. |
| $\varphi$ | $(1+\sqrt{5})/2$ | The self-similar entropy-balance point. |
| $\sqrt{\pi}$ | Boundary Gaussian normalization width | The conversion factor from pixel displacement to observation width. |
| $\alpha_{\mathrm{ext}}(P)$ | $(P-\varphi)/\sqrt{\pi}$ | The declared outer detuning coordinate. |
| $A_{\mathrm{Th}}(P)$ | $\alpha_{\mathrm{em}}^{-1}(0;P)$ | The inverse electromagnetic endpoint returned by the declared trial map. |
| $\alpha_{\mathrm{in}}(P)$ | $1/A_{\mathrm{Th}}(P)$ | The declared inner coordinate; its same-cell interpretation requires a typed bridge. |
| $E_P$ | Planck energy | The energy unit associated with the selected scale certificate. The numerical equations use Planck units unless GeV labels are attached. |
| $M_U(P)$ | $E_P \mathrm{e}^{-2\pi} P^{1/6}$ | The OPH unification-scale readout returned at the declared trial coordinate. |
| $E_{\mathrm{cell}}(P)$ | $E_P/\sqrt{P}$ | The local cell energy readout. |
| $\alpha_U(P)$ | Unified coupling solved from the heat-kernel closure equation | The coupling at $M_U(P)$ that makes the gauge representation entropy match the pixel. |
| $\alpha_i(\mu;P)$ | Gauge coupling $i=1,2,3$ at scale $\mu$ | The hypercharge, weak, and color couplings emitted by the same trial $P$. |

| Symbol | Definition | Informal meaning |
|---|---|---|
| $b_i$ | $(33/5, 1, -3)$ | The one-loop MSSM beta coefficients used as the high-scale running convention here; a declared structural selection. |
| $m_Z(P)$ | Self-consistent electroweak anchor scale | The $Z$-scale generated by $v(P)$, $\alpha_1(P)$, and $\alpha_2(P)$. |
| $v(P)$ | $E_{\mathrm{cell}}(P)\exp[-2\pi/(\beta_{\mathrm{EW}}\alpha_U(P))]$ | The electroweak transmutation scale. |
| $\beta_{\mathrm{EW}}$ | $N_c + 1 = 4$ for $N_c = 3$ | The coefficient that controls the exponential drop to the electroweak scale. |
| $A_Z(P)$ | $\alpha_{\mathrm{em}}^{-1}(m_Z^2; P)$ | The electroweak-scale electromagnetic anchor. |
| $\Delta_{\mathrm{lep}}(P)$ | Lepton transport contribution to inverse alpha | The exact one-loop kernel evaluated on the frozen phenomenological $e, \mu, \tau$ spectrum. |
| $\Delta_{\mathrm{had}}(P)$ | Hadronic transport contribution | The low-energy QCD spectral contribution needed for the pure source calculation. |
| $\Delta_{\mathrm{EW}}(P)$ | Electroweak/scheme endpoint remainder | Same-scheme finite remainder needed to connect the source anchor to the endpoint. |
| $\Delta_{\mathrm{calc}}(P)$ | Calculated charged-fermion transport contribution | The exact kernel on the supplied lepton continuation plus the screened quark continuation; not a pure source prediction. |
| $\Delta_{\mathrm{H,cal}}^{\mathrm{fp}}$ | Root-to-CODATA bookkeeping difference | The inverse-alpha amount between the naked root value and the CODATA/NIST central endpoint; it includes the comparison-pixel $\alpha_U(P_{\mathrm{C}})$ contribution and is not the post-$\alpha_U$ hadronic gap. |
| $\Delta_{\mathrm{H,req}}^{\mathrm{fp}}$ | Required hadronic endpoint correction after $A_{\alpha_U}^{\mathrm{fp}}$ | The inverse-alpha amount by which the displayed OPH output misses the CODATA/NIST central endpoint. |
| $R_Q(P)$ | $A_{\mathrm{Th}}(P) - [A_Z(P) + \Delta_{\mathrm{calc}}(P)]$ | The residual same-scheme hadronic endpoint contribution. |
| $A_{\alpha_U}^{\mathrm{fp}}$ | $\alpha_{\mathrm{root}}^{-1} + \alpha_U(P_{\mathrm{C}})$ | Mixed-provenance no-hadron comparison diagnostic; the root is source-side and $P_{\mathrm{C}}$ is CODATA-derived. |

| Symbol | Definition | Informal meaning |
|---|---|---|
| $J_{24,Q}(P)$, $\omega_Q(P)$, $\Xi_Q(P)$ | Source-side hadronic spectral objects required for a final endpoint theorem | The Jacobi/Stieltjes spectral payload, normalization, and finite same-scheme remainder that must be emitted without looking at the Thomson target. |
| $P_{\mathrm{C}}$ | CODATA/NIST comparison pixel | The pixel obtained from the CODATA/NIST Thomson endpoint under the OPH outer equation. |

## 3 Shared Collar Convention for $\chi_\nu$

The $\chi_\nu$ protected-reserve theorem uses the alpha-calibrated comparison pixel

$$P_\chi = P_{\mathrm{C}} = 1.630968209403959\ldots.$$

Its scalar reserve density is

$$\epsilon_{\mathrm{res}} = \frac{P_\chi}{24}$$

only after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass in the screen microphysics paper and in the observers library paper. This does not make $P/4$ a primitive Hilbert-space dimension. In this paper, $P/4$ is the local screen-cell entropy budget used by the electromagnetic fixed-point branch; $P_\chi/24$ is the shared scalar-reserve density used by the $\chi_\nu$ collar branch. The $P/4$ budget is a declared area reading of the cell. Native regional entropy in the executable observer grows with the number of enclosed ports rather than with crossing seams, and the physical region map that would carry an area law is a premise [11].

## 4 Step 1: The OPH Overlap Rule

OPH starts from a finite screen covered by observer patches. If $P_1$ and $P_2$ overlap, then their induced states must agree on the shared algebra:

$$\omega_{P_1}|_{\mathcal{A}(P_1\cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1\cap P_2)}. \tag{1}$$

Informally: no observer sees the whole world. Physics is what survives when neighboring local descriptions can be checked against each other and made consistent on the parts they share.

The quantitative fine-structure branch applies this rule to a single local screen cell. The cell has one outside description and one inside description. The outside description is geometric. The inside description is electromagnetic. A declared typed bridge identifies both as readings of one cell; the fixed-point theorem is conditional on that premise.

The executable observer construction runs local agreement repair on a twelve-port carrier: each native repair event moves load across one seam and records its placement coin and the transferred current [12]. Its physical identification with our Universe is unestablished.

Why this step is needed: OPH reads a dimensionless constant as a closure coordinate that finite observers locate by measurement. A declared map is then tested by comparing its certified fixed point with the measured value, and that comparison is a diagnostic.

## 5 Step 2: The Pixel Variable

Define the local pixel ratio

$$P := \frac{a_{\mathrm{cell}}}{\ell_\star^2}. \tag{2}$$

Here $a_{\mathrm{cell}}$ is the screen-cell area and $\ell_\star^2$ is the area scale emitted by the separate OPH gravity readback fixed point.

Informally: $P$ says how large the screen cell is when measured in scale-readout units. It is dimensionless, so it can be compared directly with pure numbers such as $\varphi$ and $\sqrt{\pi}$. After the gravity row emits $G$, the same scale is displayed as the Planck area in SI units. The pixel equation fixes this dimensionless ratio; it does not determine the SI scale by itself.

Why this step is needed: $\alpha$ has no units. A unitless electromagnetic number has to be matched to a unitless geometric number. Dividing the cell area by the emitted scale area gives that number without using the pixel equation to derive $G$. The executable observer construction supplies no physical area map for its screen cells, so this area reading of $P$ is a declared dictionary entry [13].

## 6 Step 3: The Golden-Ratio Balance

The self-similar balance point is

$$\varphi = 1 + \frac{1}{\varphi}, \qquad \varphi^2 - \varphi - 1 = 0, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{3}$$

Informally: $\varphi$ is the fixed point of the simplest self-similar split, where the whole-to-large ratio equals the large-to-small ratio. OPH takes it as the declared zero-detuning balance point for the local pixel.

The pixel does not sit exactly at $\varphi$. It sits at

$$\Delta_P := P - \varphi. \tag{4}$$

Informally: $\Delta_P$ is the small amount by which the realized cell is displaced from exact self-similar equilibrium. That small displacement is what becomes the electromagnetic observation strength.

Why this step is needed: a fixed point needs a reference position. The detuning principle declares the golden ratio as the self-similar reference, while $P - \varphi$ measures how far the actual cell sits away from it.

## 7 Step 4: Boundary Gaussian Normalization

The boundary normalization converts $\Delta_P$ into the outside coupling readout:

$$\alpha_{\text{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}. \tag{5}$$

Informally: the boundary Gaussian width normalizes the screen displacement. Dividing by $\sqrt{\pi}$ turns the geometric displacement into a dimensionless observation strength. The Gaussian choice belongs to the declared detuning principle.

Why this step is needed: the outside screen variable is an area displacement. The inside electromagnetic variable is a coupling. The declared boundary normalization supplies the proposed conversion between those two readings.

## 8 Step 5: The Inside Electromagnetic Readout

Let

$$A_{\text{Th}}(P) := \alpha_{\text{em}}^{-1}(0; P) \tag{6}$$

be the inverse electromagnetic coupling at zero momentum returned by the declared map at trial pixel $P$. The inside coupling is

$$\alpha_{\text{in}}(P) := \frac{1}{A_{\text{Th}}(P)}. \tag{7}$$

Informally: an observer inside the encoded world does not see $P$ as a screen area. The observer sees the strength of electromagnetism. Since particle physicists quote the low-energy coupling as an inverse number near 137, the actual coupling is $1/A_{\text{Th}}(P)$.

Why this step is needed: the measured fine-structure constant is the Thomson-limit coupling. The calculation must therefore end at zero momentum, after the electroweak anchor has been transported through all charged degrees of freedom.

## 9 Step 6: The Fixed-Point Equation

Conditional on the same-invariant bridge, closure requires the outside and inside couplings to agree:

$$\alpha_{\text{ext}}(P) = \alpha_{\text{in}}(P). \tag{8}$$

Using Eqs. (5) and (7),

$$\frac{P - \varphi}{\sqrt{\pi}} = \frac{1}{A_{\text{Th}}(P)}. \tag{9}$$

Equivalently,

$$\boxed{H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_{\text{Th}}(P)} = 0} \tag{10}$$

or

$$\boxed{P = G(P) := \varphi + \frac{\sqrt{\pi}}{A_{\text{Th}}(P)}.} \tag{11}$$

Informally: feed a trial pixel into the declared source chain. It returns a Thomson endpoint. If the bridge proves that endpoint and the detuning are two readings of one cell, closure selects a value that comes back unchanged.

Why this step is needed: a trial value is not enough. Once the bridge proves that both readings belong to one cell, self-consistency requires them to agree. That is why the conditional answer is a fixed point rather than a one-way evaluation.

## 10 Step 7: The Source Map

For a trial $P$, the source map first emits

$$M_U(P) = E_P \, \mathrm{e}^{-2\pi} \, P^{1/6}, \tag{12}$$

$$E_{\mathrm{cell}}(P) = \frac{E_P}{\sqrt{P}}. \tag{13}$$

Informally: $M_U(P)$ is the high-scale source readout, and $E_{\mathrm{cell}}(P)$ is the declared local-energy coordinate. Both are functions of the same trial input $P$. Interpreting that input as one physical screen cell requires the same-cell bridge.

Why this step is needed: the inside electromagnetic coupling has to be generated from the same $P$ used in the outside equation. These two formulas begin that inside calculation without adding a separate high-scale fit parameter.

## 11 Step 8: Electroweak Transmutation

The source branch uses

$$\beta_{\mathrm{EW}} = N_c + 1 = 4 \tag{14}$$

with $N_c = 3$. The same value has a representation-theoretic reading in the Standard Model gauge paper: the selected exterior matter package contains exactly four weak doublets, $3 + 1$ from the quark and lepton sectors of one generation, so $N_c + 1$ and the weak-doublet multiplicity name one coefficient. The identification of that abstract load with the physical screen load is the separate common-carrier receipt. The branch defines

$$v(P, \alpha_U) = E_{\mathrm{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\mathrm{EW}} \alpha_U}\right]. \tag{15}$$

Informally: the weak scale is exponentially lower than the cell scale. The unified coupling controls that descent, so changing $\alpha_U$ changes the electroweak scale sharply.

Why this step is needed: the fine-structure constant is measured at low energy, and the source map begins at the cell scale. The transmutation formula explains how the electroweak scale is produced from the same source data. After independent direct public-record closure, a positive refinement-natural identification of the screen and electroweak load carriers gives the conditional bridge $N_{\mathrm{bridge}} = \pi \exp[6\pi/(P\alpha_U)]$, certified by interval arithmetic at $3.53 \times 10^{122}$. The frozen FZ-01 capacity registration carries this uncorrected bridge value, $3.532131543 \times 10^{122}$, as the null alternative to its band $[3.30836, 3.31137] \times 10^{122}$ [9]. The bridge does not construct the public-record readback.

## 12 Step 9: One-Loop Gauge Running

For $i = 1, 2, 3$, define

$$(b_1, b_2, b_3) = \left(\frac{33}{5},\, 1,\, -3\right), \tag{16}$$

and run the couplings by

$$\alpha_i^{-1}(\mu; P, \alpha_U) = \alpha_U^{-1} + \frac{b_i}{2\pi} \log\!\left(\frac{M_U(P)}{\mu}\right). \tag{17}$$

The triple $(33/5, 1, -3)$ is the one-loop MSSM beta-coefficient set. Its use is a declared structural selection: the Standard Model coefficient triple has no pixel-residual root in the declared scan, no supersymmetric threshold matching enters the map, and the convention is a counted structural selection.

Informally: once $\alpha_U$ is guessed, all three gauge couplings at lower scales are fixed. The same high-scale coupling determines $\alpha_1$, $\alpha_2$, and $\alpha_3$.

Why this step is needed: the electromagnetic coupling is a mixture of the weak and hypercharge couplings. Running the gauge couplings supplies the ingredients needed to form that mixture at the $Z$-scale.

## 13 Step 10: The Self-Consistent $Z$-Scale

The hypercharge coupling in Standard Model normalization is

$$\alpha_Y(\mu; P) = \frac{3}{5}\alpha_1(\mu; P). \tag{18}$$

The tree-level $Z$-mass readout is

$$m_Z(\mu; P, \alpha_U) = \frac{v(P, \alpha_U)}{2}\sqrt{4\pi\alpha_2(\mu; P, \alpha_U) + 4\pi\alpha_Y(\mu; P, \alpha_U)}. \tag{19}$$

The source point uses the self-consistency condition

$$\mu = m_Z(\mu; P, \alpha_U). \tag{20}$$

Informally: the $Z$-scale is determined inside the calculation. It is the scale at which the running couplings and the electroweak transmutation formula reproduce their own $Z$-mass readout.

Why this step is needed: using a measured $m_Z$ here would hide experimental input inside the source map. The self-consistency condition keeps the source calculation closed.

## 14 Step 11: Heat-Kernel Gauge Closure

For SU(2), the irreducible representations are labeled by $j = n/2$, $n = 0, 1, \ldots, N_2$, with

$$d_j = 2j + 1, \qquad C_2(j) = j(j+1). \tag{21}$$

For SU(3), representations are labeled by highest weights $(p, q)$, $0 \le p, q \le N_3$, with

$$d_{p,q} = \frac{(p+1)(q+1)(p+q+2)}{2}, \tag{22}$$

$$C_2(p,q) = \frac{p^2+q^2+pq+3p+3q}{3}. \tag{23}$$

For a compact group $G$ in this finite representation cutoff, define

$$Z_G(t) = \sum_R d_R \, \mathrm{e}^{-tC_2(R)}, \tag{24}$$

$$\bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R \, \mathrm{e}^{-tC_2(R)} \log d_R. \tag{25}$$

The displayed evaluations use the representation cutoffs $N_2 = 120$ and $N_3 = 90$. At the closure arguments below, $t_2 \approx 1.3$ and $t_3 \approx 4.7$, the first omitted terms carry weights below $\mathrm{e}^{-4000}$, so the cutoffs are inert at every displayed precision, and the interval certificate bounds the tails beyond any finite cutoff.

The heat-kernel source parameters are

$$t_2 = 4\pi^2 \alpha_2(m_Z; P, \alpha_U), \qquad t_3 = 4\pi^2 \alpha_3(m_Z; P, \alpha_U). \tag{26}$$

The pixel-closure equation is

$$\boxed{\bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3) = \frac{P}{4}.} \tag{27}$$

For a trial $P$, Eq. (27) is solved for $\alpha_U(P)$. The closure sums the SU(2) and SU(3) representation towers only; the selection that hypercharge carries no representation entropy against the $P/4$ budget is a declared choice. The repair round count used by the finite current construction counts the $\mathfrak{u}(1)$ generator ($8 + 3 + 1 = 12$); the two conventions are distinct declared choices.

Informally: the representation entropy carried by the weak and color sectors must match the pixel capacity assigned to the cell. This step is what locks the unified coupling to the same $P$ that appears in the outer equation.

Why this step is needed: $\alpha_U$ should not be chosen by hand. The heat-kernel closure turns the finite representation content of the gauge sectors into an equation for $\alpha_U(P)$.

## 15 Step 12: The Electroweak Anchor

Once $\alpha_U(P)$ and $m_Z(P)$ are solved, define

$$\alpha_Y(m_Z; P) = \frac{3}{5}\alpha_1(m_Z; P), \tag{28}$$

$$\alpha_{\mathrm{em}}(m_Z^2; P) = \left( \frac{1}{\alpha_2(m_Z; P)} + \frac{1}{\alpha_Y(m_Z; P)} \right)^{-1}. \tag{29}$$

The source-locked electroweak anchor is

$$\boxed{A_Z(P) := \alpha_{\mathrm{em}}^{-1}(m_Z^2; P).} \tag{30}$$

The weak mixing readout is

$$\sin^2\theta_W(m_Z;P)=\frac{\alpha_{\rm em}(m_Z^2;P)}{\alpha_2(m_Z;P)}. \tag{31}$$

Informally: the source map has reached the electromagnetic coupling at the electroweak anchor scale. The low-energy 1/137 number appears only after this anchor is transported to zero momentum.

Why this step is needed: $A_Z(P)$ is the clean place where the source map meets standard electroweak physics. From here the problem becomes a transport problem for the electromagnetic current.

# 16 Step 13: Frozen Charged-Spectrum Continuation Used by the Transport

The one-loop kernel below is exact for supplied fermion masses, but the charged spectrum supplied to it in this section is a phenomenological continuation, not an OPH-derived mass theorem. It uses $N_c=3$, $N_g=3$, and the additional assumptions

$$\epsilon=\frac{1}{6},\qquad \delta=\frac{\beta_{\rm EW}}{2N_cN_g}=\frac{2}{9}. \tag{32}$$

No physical charged-spectrum derivation is claimed here; such a derivation requires a source-to-mass attachment, a phase-transport law fixing $\delta=2/9$, and an attachment of the ordered roots to physical charged-family lines. For a positive-semidefinite square-root-mass carrier

$$C=aI+\rho(e^{i\delta}R+e^{-i\delta}R^2),\qquad Q=\frac{1+2(\rho/a)^2}{3},$$

so $\rho/a=1/\sqrt{2}$ is exactly equivalent to the empirical Koide value $Q=2/3$. At balance this identity holds throughout the positive chamber $|\delta|\le\pi/12 \pmod{2\pi/3}$, and it does not select the phase. The balance is a target-informed premise and the phase $\delta$ is supplied. The finite tracial construction that realizes the balance, and the maps a physical charged-lepton reading would additionally require, are stated in the Koide paper [5]. Define three Koide roots

$$r_k=1+\sqrt{2}\cos\!\left(\delta+\frac{2\pi k}{3}\right),\qquad k=0,1,2, \tag{33}$$

then sort them in increasing order and write the sorted list as $(r_1,r_2,r_3)$.

The quark exponent vectors are

$$\boldsymbol{n}_u=(2N_c,N_c,0)=(6,3,0),\qquad \boldsymbol{n}_d=(2N_c,N_c+1,N_c-1)=(6,4,2). \tag{34}$$

With $v=v(P)$,

$$m_u=\frac{v}{\sqrt{2}}\epsilon^6,\qquad m_c=\frac{v}{\sqrt{2}}\epsilon^3,\qquad m_t=\frac{v}{\sqrt{2}}, \tag{35}$$

$$m_d=\frac{v}{\sqrt{2}}\epsilon^6,\qquad m_s=\frac{v}{\sqrt{2}}\epsilon^4,\qquad m_b=\frac{v}{\sqrt{2}}\epsilon^2. \tag{36}$$

For charged leptons the exponent vector is

$$\boldsymbol{n}_e=(7,4,3). \tag{37}$$

This vector and the normalization below are declared phenomenological transport inputs. The declared premises do not derive $\boldsymbol{n}_e = (7,4,3)$ or the resulting determinant identity $\det M_e = v(P)^3/(2\,6^{14})$. The electron entry is selected by a finite scan: with $n_\tau = N_g = 3$ and $n_\mu = N_g+1 = 4$ fixed by the generation count, the electron exponent runs over the integers 5 to 11, and the entry with the smallest maximal pairwise logarithmic residual between the squared balanced-circulant roots at $\delta = (N_c+1)/(2N_cN_g)$ and the corresponding powers of $\epsilon = 1/6$ is kept; the minimizer is 7. The scan compares the declared circulant roots with the $\epsilon$ ladder and reads no measured mass, and the vector is a declared transport input all the same. Every common shift $(7+k, 4+k, 3+k)$ has the same ratio residual while changing the determinant. The extra factor $2^{1/6}$ below is asserted rather than derived. Define

$$\log g_c = \frac{1}{3}\sum_{a=1}^{3} \log\left(\frac{r_a^2\sqrt{2}\,6^{n_{e,a}}}{v}\right), \qquad s_0 = \mathrm{e}^{-\log g_c}, \qquad s_e = s_0\,2^{1/6}. \tag{38}$$

Then

$$m_e = s_e r_1^2, \qquad m_\mu = s_e r_2^2, \qquad m_\tau = s_e r_3^2. \tag{39}$$

Equations (38)–(39) define a transport input, not a source-derived charged-lepton mass law. No charged RG/threshold theorem maps this coordinate to physical pole masses.

Informally: this section specifies a phenomenological spectrum used by the transport diagnostic. The lepton vacuum-polarization kernel is an exact one-loop calculation conditional on those masses; that exactness does not determine the masses themselves. The quark part is also a perturbative continuation. The confined hadronic QCD spectral measure is a separate low-energy object.

Why this step is needed: vacuum polarization depends on charged particles. The transport diagnostic cannot be evaluated until a charged spectrum and its charges are supplied. A pure OPH source theorem would first have to replace the continuation spectrum by independently derived charged masses.

## 17 Step 14: Exact One-Loop Fermion Transport

For a fermion of mass $m_f$, electric charge $Q_f$, and multiplicity $N_f$, define

$$K_f(Q^2; m_f, Q_f, N_f) = \frac{2N_fQ_f^2}{\pi}\int_0^1 x(1-x)\log\left(1+\frac{Q^2x(1-x)}{m_f^2}\right)\mathrm{d}x. \tag{40}$$

The integral has the following closed form. Let

$$z = \frac{Q^2}{m_f^2}, \qquad a = \frac{z}{4}. \tag{41}$$

Then

$$\int_0^1 x(1-x)\log(1+zx(1-x))\,\mathrm{d}x = -\frac{5}{18} + \frac{1}{6a} + \frac{(2a-1)\sqrt{1+a}\,\operatorname{asinh}(\sqrt{a})}{6a^{3/2}}, \tag{42}$$

with

$$\operatorname{asinh}(\sqrt{a}) = \log(\sqrt{a}+\sqrt{1+a}). \tag{43}$$

The lepton contribution is

$$\Delta_{\mathrm{lep}}(P) = K_e(m_Z(P)^2; m_e, 1, 1) + K_\mu(m_Z(P)^2; m_\mu, 1, 1) + K_\tau(m_Z(P)^2; m_\tau, 1, 1). \tag{44}$$

The naive five-quark contribution is

$$\begin{aligned}\Delta_q^{\mathrm{naive}}(P) = {} & K_u(m_Z(P)^2; m_u, 2/3, 3) + K_d(m_Z(P)^2; m_d, -1/3, 3) \\ & + K_s(m_Z(P)^2; m_s, -1/3, 3) + K_c(m_Z(P)^2; m_c, 2/3, 3) + K_b(m_Z(P)^2; m_b, -1/3, 3).\end{aligned} \tag{45}$$

The declared hadronic screening ansatz is

$$S_{\mathrm{calc}}(P) = 1 - \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{46}$$

The form of $S_{\mathrm{calc}}$ is a declared selection of the quark continuation and carries no perturbative derivation: the leading perturbative correction to quark vacuum polarization is the enhancement $R \to R_0(1 + \alpha_s/\pi + \cdots)$ with no $N_c$ factor, about $+3.7$ percent at $m_Z$, while $S_{\mathrm{calc}} \approx 0.887$ is an 11 percent suppression standing in for confinement cutting off the light-quark logarithms. Thus

$$\Delta_{\mathrm{calc}}(P) = \Delta_{\mathrm{lep}}(P) + S_{\mathrm{calc}}(P)\Delta_q^{\mathrm{naive}}(P). \tag{47}$$

Informally: the supplied charged-lepton continuation is propagated through an exact one-loop kernel. Quarks are treated by the one-loop kernel times the declared screening ansatz. The expected gap is the confined hadronic spectral transport plus the matching terms needed to use the same electromagnetic-current convention all the way to the endpoint.

Why this step is needed: the Thomson endpoint is not the same as the $Z$-scale anchor. Charged particles screen the electromagnetic current between those scales, and the kernel is the mathematical form of that screening.

## 18 Step 15: The Source Calculation Endpoint

The calculated endpoint is

$$A_{\mathrm{calc}}(P) = A_Z(P) + \Delta_{\mathrm{calc}}(P). \tag{48}$$

Putting $A_{\mathrm{calc}}$ into Eq. (11) gives the source map

$$G_{\mathrm{calc}}(P) = \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{calc}}(P)}. \tag{49}$$

The numerical solve uses

$$G_{\mathrm{calc}}(P_{\mathrm{root}}) = P_{\mathrm{root}} \tag{50}$$

and emits

$$P_{\mathrm{root}} = 1.630972095858897376964513903506955628479\ldots, \tag{51}$$

$$\alpha_{\mathrm{root}}^{-1} = 136.994835177412937295289429464436902857 6\ldots. \tag{52}$$

The fixed point is interval-certified as existing and unique on its interval (interval certificate, available in the public artifact bundle [4]; enclosure widths $6.8 \times 10^{-28}$ in $P$ and $7.2 \times 10^{-24}$ in $\alpha^{-1}$, Lipschitz bound $L \leq 0.0724$). The source anchor at the certified root point is

$$A_Z(P_{\mathrm{root}}) = 128.308268057987597347904057614084758875033 1\ldots. \tag{53}$$

The calculated transport contribution is

$$\Delta_{\mathrm{calc}}(P_{\mathrm{root}}) = 8.686567119425339947385371850352143982 52\ldots. \tag{54}$$

Informally: the fixed-point algebra lands at a stable value near 137, with no measured alpha inserted into the solve. Its closeness to the measured value depends on the declared screening ansatz; with the measured hadronic payload in its place the endpoint is 136.3827548 (Section 22). The numerical interval does not bound the missing physical contributions.

Why this step is needed: this is the non-circular check. It shows what the OPH source chain produces before the calibrated hadronic endpoint correction is added.

The P-dependent detector of the executable observer construction takes this root as its closure input. Under a chosen charge assignment it sets $g(P_{\mathrm{root}}) = \sqrt{4\pi\alpha(P_{\mathrm{root}})} = 0.3028676131\ldots$ and records a binary detector observation whose probabilities depend on $P$ and which feeds the next native repair; that use leaves the endpoint status of the root unchanged [10].

## 19 Step 16: The CODATA/NIST Measured Endpoint and Comparison Pixel

The CODATA/NIST 2022 inverse fine-structure constant [26, 28] is

$$A_{\mathrm{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{55}$$

with concise form 137.035999177(21). The corresponding comparison pixel is

$$P_{\mathrm{C}} = \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{C}}} = \begin{array}{l}\texttt{1.630968209403959324879279847782648941335982851627925060666 1}\\ \texttt{507533907793398933432}\end{array} \tag{56}$$

The CODATA/NIST central coupling is

$$\alpha(0) = \frac{1}{A_{\mathrm{C}}} = \begin{array}{l}\texttt{0.007297352564331425030245795264691683228066021313365360495 7}\\ \texttt{798803819933561573639928}\end{array} \tag{57}$$

Informally: once the measured Thomson endpoint is supplied, the outer OPH equation fixes the corresponding comparison pixel immediately. It is the direct readout of one equation.

Why this step is needed: the measured endpoint and the comparison pixel are two algebraic forms of the same comparison definition. Reporting both makes the measured provenance of the pixel explicit; it does not identify that coordinate with a source-selected screen cell.

**Two-$P$ provenance audit.** There are two distinct pixel numerals in this paper. The source-only solve gives $P_{\mathrm{root}}$ in Eq. (51); it is the branch that audits what the OPH source chain emits before a same-scheme hadronic endpoint correction is supplied. The comparison value $P_{\mathrm{C}}$ in Eq. (56) is different by about $3.9 \times 10^{-6}$ in $P$-space because it is defined from the measured CODATA/NIST Thomson endpoint. Thus the published comparison pixel is a measured-endpoint display, not a derivation of $\alpha$. Downstream quantities that use $P_{\mathrm{C}}$ inherit that measured endpoint by definition; downstream quantities that use $P_{\mathrm{root}}$ are source-branch diagnostics. The registered conditional primordial tilt $n_s = 1 - P/48 = 0.9660214956\ldots$ is one such downstream quantity; it is evaluated at $P_{\mathrm{C}}$ [7, 9].

## 20 Step 17: Endpoint Accounting at the CODATA/NIST Comparison Pixel

At $P = P_{\mathrm{C}}$, the source point gives

$$A_Z(P_{\mathrm{C}}) = \frac{128.307965473286248209961108741756716187245476180365356460053421696351177841682856440787247 28}{} \tag{58}$$

$$\Delta_{\mathrm{calc}}(P_{\mathrm{C}}) = \frac{8.6865678427085284009854425428859697682672217376487364233784}{577389993784459106783080961179590} \tag{59}$$

These are non-certified artifact displays: the quoted digit strings are the run's output at its declared convergence budget, and independent recomputation agrees with $A_Z(P_{\mathrm{C}})$ to 27 significant digits rather than to the full displayed precision; the certified interval artifacts are unaffected and land inside their intervals far deeper. The transport required by the CODATA/NIST measured endpoint is

$$\Delta_{\mathrm{req}}(P_{\mathrm{C}}) = A_{\mathrm{C}} - A_Z(P_{\mathrm{C}}) = \frac{8.728033703713751790038891258243283812754523819634643539946578303648822158317143559212752 72}{} \tag{60}$$

Therefore the source-side residual at the CODATA/NIST comparison pixel is

$$\begin{aligned} R_Q(P_{\mathrm{C}}) &= \Delta_{\mathrm{req}}(P_{\mathrm{C}}) - \Delta_{\mathrm{calc}}(P_{\mathrm{C}}) \\ &= \frac{0.041465861005223389053448715357314044487302081985907116568120564649443712406465251116634 76}{} \end{aligned} \tag{61}$$

Informally: the CODATA/NIST measured value differs from the pure source calculation by a small, precisely located inverse-alpha amount. Neither declared map computes same-scheme low-energy hadronic transport to the Thomson endpoint, and inserting the measured hadronic payload widens the difference rather than closing it (Section 22). The fixed-point solve and this accounting do not identify the physical source of the gap [6, 8].

Why this step is needed: this accounting prevents the hadronic contribution from being hidden inside the final number. It shows the source anchor, the calculated transport, the required transport, and the residual in the same units.

## 21 Step 18: The OPH Output and the Required Hadronic Correction

The fixed-point calculation emits

$$A_{\mathrm{calc}}(P_{\mathrm{root}}) = \alpha_{\mathrm{root}}^{-1} = \frac{136.9948351774129372952894294644369028575612061510353931845 02}{}. \tag{62}$$

The interval certificate and continuous-integration identity test are included in the public artifact bundle [4]. At the CODATA-derived comparison pixel, the finite-screen unified gauge-width contribution is

$$\alpha_U(P_{\mathrm{C}}) = 0.041124336195630495.$$

Combining that comparison-pixel gauge width with Equation (62) gives the mixed-provenance nohadron diagnostic

$$A^{\mathrm{fp}}_{\alpha_U} = A_{\mathrm{calc}}(P_{\mathrm{root}}) + \alpha_U(P_{\mathrm{C}}) = 137.035959513608567790289429464437\ldots, \tag{63}$$

or

$$\alpha^{\mathrm{fp}}_{\mathrm{src}+U} = (A^{\mathrm{fp}}_{\alpha_U})^{-1} = 0.007297354676461352132872759596 57\ldots. \tag{64}$$

This $A^{\mathrm{fp}}_{\alpha_U}$ value is comparison bookkeeping, not a source-only prediction: $\alpha^{-1}_{\mathrm{root}}$ is evaluated on the source root, while $\alpha_U(P_{\mathrm{C}})$ inherits the measured endpoint through $P_{\mathrm{C}}$, and the sum is not a fixed point of any single declared map.

Compared with the CODATA/NIST central measured endpoint,

$$A_{\mathrm{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{65}$$

the CODATA-minus-diagnostic bookkeeping gap is

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = A_{\mathrm{C}} - A^{\mathrm{fp}}_{\alpha_U} = \texttt{0.000039663391432209710570535563}\ldots. \tag{66}$$

Equivalently,

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = 0.0000396634 \pm 0.000000021, \qquad \frac{\Delta^{\mathrm{fp}}_{\mathrm{H,req}}}{A_{\mathrm{C}}} = 2.89438 \times 10^{-7}. \tag{67}$$

This is the inverse-alpha gap between the mixed diagnostic and the CODATA/NIST central value. It is the gap left by the declared screening ansatz; inserting the measured hadronic vacuum polarization instead does not close it (Section 22). The subtraction neither calculates same-scheme transport of the Ward-projected electromagnetic current nor identifies the physical source of the gap. The $2.89438 \times 10^{-7}$ ratio describes the mixed diagnostic only. The distinct self-consistent gauge-width map has fixed point $\alpha^{-1} = 137.035660136946577\ldots$ and relative residual $2.5 \times 10^{-6}$, about $1.6 \times 10^4$ measurement sigma.

For bookkeeping, the full root-to-CODATA difference is

$$\begin{aligned}\Delta^{\mathrm{fp}}_{\mathrm{H,cal}} &= A_{\mathrm{C}} - A_{\mathrm{calc}}(P_{\mathrm{root}}) \\ &= \texttt{0.041163999587062704710570535563097142439} \\ &= \alpha_U(P_{\mathrm{C}}) + \Delta^{\mathrm{fp}}_{\mathrm{H,req}}.\end{aligned} \tag{68}$$

This larger difference is bookkeeping: it contains the comparison-pixel unified gauge-width contribution and the CODATA-minus-diagnostic gap.

Equivalently, the CODATA/NIST central value can be written as

$$A_{\mathrm{C}} = \alpha^{-1}_{\mathrm{root}} + \alpha_U(P_{\mathrm{C}})\, C_{24,Q}, \qquad C_{24,Q} = 1.000964474933857388040135711404515 2\ldots. \tag{69}$$

The factor $C_{24,Q}$ is a back-solved calibration: compact comparison accounting, solved from the measured endpoint, for the finite same-scheme hadronic endpoint transport needed for source-only closure. Equivalently, $\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = \alpha_U(P_{\mathrm{C}})(C_{24,Q} - 1)$. The factor is not a source-emitted hadronic/QCD payload.

Here $A_{\mathrm{C}}$ is the CODATA/NIST 2022 inverse fine-structure constant [26, 27]. Closing the endpoint gap requires a source-derived hadronic spectral calculation, which this paper does not supply.

The endpoint decomposition is

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P). \tag{70}$$

The calculated transport contains the exact one-loop charged-lepton contribution and a structured quark-screening continuation. The residual pure source object can be written compactly as

$$R_Q(P) = \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P) - [\Delta_{\mathrm{calc}}(P) - \Delta_{\mathrm{lep}}(P)], \tag{71}$$

where the right side is understood in the same renormalization and endpoint scheme as $A_Z(P)$.

In the screening notation used by the endpoint calculation, let

$$x(P) = \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{72}$$

At $P = P_{\mathrm{C}}$,

$$S_{\mathrm{required}} = \frac{\texttt{0.8954001326476587978058002831816706413077098648122916484483}}{\texttt{0591011545242521273086888107864576}} \tag{73}$$

$$c_Q = \frac{\texttt{0.6580257599271554356382301702323600504249200990705863960856}}{\texttt{6007832470711257322011342309305088}} \tag{74}$$

where

$$S_{\mathrm{required}} = 1 - x + c_Q x^2. \tag{75}$$

Informally: the needed hadronic and endpoint correction can be summarized as a small second-order screening coefficient. A pure source proof has to emit this coefficient from a Ward-projected hadronic spectral measure.

Why this step is needed: hadrons are not pointlike free quarks at low energy. The coefficient records the screening that the CODATA/NIST endpoint requires in this endpoint convention; it is back-solved from that endpoint.

## 22 Step 19: Why Raw PDG $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)$ Is a Different Quantity

The direct PDG/CERN diagnostic uses

$$A_Z(P_{\mathrm{C}}) = \frac{\texttt{128.30796547328624820996110874175671618724547618036535646005}}{\texttt{342169635117784168285644}} \tag{76}$$

$$\Delta_{\mathrm{lep}}(P_{\mathrm{C}}) = \frac{\texttt{4.30939786645220402713174389753448940184871566055767731947115}}{\texttt{2808966568031325790646612 9}} \tag{77}$$

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.02761. \tag{78}$$

It forms

$$A_L = A_Z + \Delta_{\mathrm{lep}}, \qquad A_{\mathrm{PDGdiag}} = \frac{A_L}{1 - \Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)}. \tag{79}$$

Numerically,

$$A_{\mathrm{PDGdiag}} = \begin{array}{l}\texttt{136.382895072695577121415124218977165118002233508081154453995}\\ \texttt{950072020253794568912379458128 9}\end{array} \tag{80}$$

This differs from 137.035999177. Holding this raw hadronic denominator shift fixed, the target would require

$$\Delta\alpha_{\mathrm{had,req}} = \begin{array}{l}\texttt{0.032244343557887288822268499236957947422007347612226081506 5}\\ \texttt{9308922187138092611795014143854 9}\end{array} \tag{81}$$

or a same-scheme source-anchor bridge of

$$\begin{array}{c}(\Delta\alpha_{\mathrm{had,req}} - \Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)) \cdot \alpha^{-1}(0) = \\ \texttt{0.635071899984577762907147360708794410905808159076966220475 4}\\ \texttt{2549468225412699135291338 71}\end{array} \tag{82}$$

inverse-alpha units, with $\Delta\alpha_{\mathrm{had,req}}$ from Eq. (81), $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.02761$, and $\alpha^{-1}(0) = 137.035999177$ the CODATA central endpoint. With the pinned KNT19 payload $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = 0.027609 \pm 0.000112$, the same construction gives $\alpha^{-1}(0) = 136.3827548$, inside the interval from 136.3670481 to 136.3984652. The measured value lies outside this interval. The corresponding same-scheme anchor gap lies in $[0.6199, 0.6506]$ inverse-alpha units and contains the standard on-shell reference deficit 0.6309; both numbers measure the same anchor gap, computed once with the OPH lepton transport and once with standard running, so their agreement checks the bookkeeping and does not make the endpoint compatible with the measured value. This compare-only row does not identify the physical source of the gap [8].

Informally: the raw electroweak-review hadronic running number is a useful diagnostic. The OPH endpoint needs the same electromagnetic current and the same inverse-alpha endpoint convention used by $A_Z(P)$. Mixing the two conventions moves the answer by a visible amount.

Why this step is needed: a common mistake is to insert a published $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)$ number as though it were the OPH endpoint contribution. This section shows the numerical consequence of that convention mismatch.

## 23 Step 20: The Measured Hadronic Spectral Input

The standard measured hadronic vacuum-polarization relation uses

$$\Delta\alpha_{\mathrm{had}}(q^2) = -\frac{\alpha q^2}{3\pi}\,\mathrm{P.V.}\int \frac{R(s)}{s(s-q^2)}\,\mathrm{d}s, \tag{83}$$

where $R(s)$ is the measured ratio of the bare $e^+e^- \to$ hadrons cross section to the pointlike $e^+e^- \to \mu^+\mu^-$ cross section.

The independently documented measured quantity in the standard electroweak literature is usually quoted as $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2)$. It is a dimensionless change in the running electromagnetic coupling at the $Z$-boson mass, not the inverse-alpha correction $\Delta^{\mathrm{fp}}_{\mathrm{H,req}}$ used in Eq. (66). For example, Davier, Hoecker, Malaescu, and Zhang report

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (275.7 \pm 1.0) \times 10^{-4}$$

from $e^+e^-$-based data [20]. A perturbative update by Erler and Ferro-Hernandez gives

$$\Delta\alpha^{(5)}_{\rm had}(M_Z^2) = (276.29 \pm 0.38 \pm 0.62) \times 10^{-4}$$

when low-energy cross-section data are used as input [23].

The comparison residual in Eq. (66) is different. It is expressed in inverse-alpha units and in the endpoint convention used by $A_Z(P)$. The standard published hadronic-running values document the physics source of the correction. They do not by themselves give the OPH same-scheme inverse-alpha correction to arbitrary precision. The community does publish the data machinery needed for this kind of calculation. Jegerlehner's alphaQED package provides hadronic running routines, covariance data, and an integration routine for custom kernels [25]. That is the right empirical route once the OPH endpoint kernel and finite same-scheme remainder are fixed.

In the OPH endpoint convention this is represented as a same-current spectral functional,

$$\Delta_{\rm had}(P) = \frac{m_Z(P)^2}{3\pi} \int \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,{\rm d}s, \tag{84}$$

plus the same-scheme finite remainder needed by Eq. (70).

Informally: the measured hadronic input supplies the electromagnetic spectral information required for the endpoint. The single published number 0.0276 is one weighted integral of that spectral information. The OPH endpoint needs a different weighted integral plus the same-scheme finite remainder. A pure source theorem requires the same spectral object from OPH source data.

Why this step is needed: the dispersion relation explains why measured $e^+e^- \to$ hadrons data are the right empirical input. They measure the electromagnetic spectral function that the endpoint transport requires.

## 24 No-Go: The 24-Slot Register Scale Does Not Determine the Hadronic Residual

**Theorem 1** (Coarse 24-slot register data do not determine the Thomson residual)**.** *The specified OPH source objects $P$, $\alpha_U(P)$, $N_c = 3$, $N_g = 3$, and the 24-slot repair-register count do not determine the exact low-energy Thomson residual.*

*Proof.* The hadronic endpoint contribution depends on the Ward-projected spectral functional

$$\Delta_{\rm had}(P) = \frac{m_Z(P)^2}{3\pi} \int_0^\infty \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,{\rm d}s \tag{85}$$

up to the same-scheme finite endpoint remainder. Let

$$K_P(s) = \frac{m_Z(P)^2}{3\pi s[s+m_Z(P)^2]}.$$

This kernel is not constant. Two positive spectral measures can therefore have the same coarse register count, the same leading normalization, and the same electroweak branch data while giving different residuals. For example, with $s_1 \neq s_2$,

$$\rho_1(s) = M\delta(s-s_1), \qquad \rho_2(s) = M\delta(s-s_2)$$

have equal total mass, but

$$\int K_P(s)\rho_1(s)\,\mathrm{d}s = MK_P(s_1), \qquad \int K_P(s)\rho_2(s)\,\mathrm{d}s = MK_P(s_2),$$

which are unequal because $K_P$ is strictly decreasing on positive support. Thus the register count does not select the spectral measure or the residual. Even at fixed total mass $M$, moving a positive atom toward zero or toward infinity makes its moment arbitrarily large or small. A quantitative scale bound needs additional support or moment assumptions. □

The hadronic precision audit sharpens the no-go statement. The $\rho_Q(s;P)$ object used here is the two-current marginal of a larger source-derived hadronic spectral backend: a source QCD quotient ensemble, source parameter map, Ward-normalized current data, Stieltjes/Jacobi export, same-scheme remainder, and systematics/no-target-leak receipt bundle. That marginal can close the running-$\alpha$/HVP transport only after it is emitted by the backend. It cannot be recycled as a full hadronic precision source, because HLbL and rare decay long-distance amplitudes require four-current and transition spectral data from the same source law.

**Theorem 2** (Tautological 24-Jacobi realization)**.** *For any positive residual $R > 0$, there is an exact 24-point Jacobi/Stieltjes representation that reproduces $R$. Such a representation is non-predictive unless its nodes, weights, normalization, and finite remainder are emitted by OPH source rules before comparison with the Thomson endpoint.*

*Proof.* Fix $m > 0$. Choose pairwise distinct positive spectral nodes $s_1, \dots, s_{24} > 0$ and positive normalized weights $u_j > 0$ with $\sum_j u_j = 1$. The finite inverse Stieltjes/Favard construction gives a positive $24 \times 24$ Jacobi matrix $J_{24,Q}$ whose $e_1$-spectral measure is

$$\mathrm{d}\mu_0(s) = \sum_{j=1}^{24} u_j \delta(s - s_j).$$

Equivalently,

$$e_1^\top f(J_{24,Q}) e_1 = \sum_{j=1}^{24} u_j f(s_j)$$

for every function $f$ on the spectrum. Define

$$D(J_{24,Q}; m) = e_1^\top \left[ J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1.$$

Since $s_j > 0$,

$$D(J_{24,Q}; m) = \sum_{j=1}^{24} u_j \left( \frac{1}{s_j} - \frac{1}{s_j + m^2} \right) > 0.$$

For any same-scheme finite remainder $\Xi_Q < R$, set

$$\omega_Q = \frac{3\pi(R - \Xi_Q)}{D(J_{24,Q}; m)}.$$

Then

$$\frac{\omega_Q}{3\pi} e_1^\top \left[ J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1 + \Xi_Q = R.$$

Distinctness makes the moment inner product nondegenerate through degree 23, so the finite Jacobi construction has 24 sites with positive off-diagonals. Repeated nodes instead collapse the support and do not justify a 24-site irreducible Jacobi matrix. The construction reproduces the chosen residual exactly, but the freedom in $s_j$, $u_j$, $\omega_Q$, and $\Xi_Q$ means it is a back-solving theorem rather than a source derivation. □

The source-side object needed for a final OPH endpoint theorem is therefore

$$R_Q(P) = \frac{\omega_Q(P)}{3\pi} e_1^\top \left[ J_{24,Q}(P)^{-1} - \left( J_{24,Q}(P) + m_Z(P)^2 I \right)^{-1} \right] e_1 + \Xi_Q(P), \tag{86}$$

or equivalently the source-emitted spectral density $\rho_Q(s; P)$ plus the same-scheme finite remainder. The required dependency graph has no path from the measured Thomson endpoint into $J_{24,Q}$, $\omega_Q$, $\Xi_Q$, or $\rho_Q$. The available execution embeds target constants, uses a mismatched $P$, conflates the closure residual with the total payload, and supplies a sampled envelope rather than a certified interval. It does not support a physical conclusion. A qualifying contract fixes one coordinate schema, requires a source-emitted function over the same $P$, certifies quadrature and interval bounds, and scores a detached artifact. The local source-payload validator checks declared contract consistency; it does not replay the named backend proof or establish spectral provenance. An arbitrary nonempty certificate label cannot establish those premises; in particular, a payload whose anchor and other components sum to an interval disjoint from its declared total endpoint is inconsistent. Such inputs are rejected, and a consistent but unreplayed contract is non-promoting. The separate executable interval certificates for the two incomplete declared maps retain their stated mathematical scope.

## 25 Step 21: The Conditional Pure Source Theorem

**Theorem 3** (OPH fine-structure endpoint, conditional pure source form)**.** *Assume:*

*(i) the OPH overlap-consistency branch emits the source map $P \mapsto A_Z(P)$ by Eqs. (12)–(30);*

*(ii) the Ward-projected $\mathrm{U}(1)_Q$ transport theorem emits a same-scheme endpoint map*

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P);$$

*(iii) a target-independent source law selects this source and transport map from the admitted map family;*

*(iv) a typed same-quantity bridge identifies $(P-\varphi)/\sqrt{\pi}$ and $1/A_{\mathrm{Th}}(P)$ as readings of one physical screen cell, with $1/A_{\mathrm{Th}}(P)$ identified as its Thomson-limit electromagnetic coupling;*

*(v) $G(P) = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ is a self-map and a contraction on a nonempty closed branch interval $I \subset \mathbb{R}$;*

*(vi) the interval image contains the root and the residual bound is certified.*

*Then there is a unique $P_\star \in I$ satisfying $G(P_\star) = P_\star$, and the fine-structure constant on that branch is*

$$\alpha(0) = \frac{1}{A_{\mathrm{Th}}(P_\star)} = \frac{P_\star - \varphi}{\sqrt{\pi}}.$$

*Proof.* By assumption (v), $I$ is nonempty and complete in the Euclidean metric, and $G : I \to I$ is a contraction. Banach's fixed-point theorem gives a unique $P_\star \in I$ such that $G(P_\star) = P_\star$. Equation (11) gives

$$P_\star - \varphi = \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P_\star)}.$$

Dividing by $\sqrt{\pi}$ gives

$$\frac{P_\star - \varphi}{\sqrt{\pi}} = \frac{1}{A_{\mathrm{Th}}(P_\star)}.$$

The left side is $\alpha_{\mathrm{ext}}(P_\star)$, and the right side is $\alpha_{\mathrm{in}}(P_\star)$. Assumptions (iii)–(iv) select this map and identify their common value with the Thomson-limit electromagnetic coupling $\alpha(0)$. □

Informally: no source-derived hadronic spectral endpoint map supplies the required transport. A physical fine-structure conclusion also requires target-independent map selection and the same-quantity bridge. With those premises and the interval proof, the selected full source map has one fixed point and the bridge identifies its value with the fine-structure constant.

The interval form of assumptions (v)–(vi) is direct: on any nonempty closed interval $I$ where interval-arithmetic evaluation certifies $G(I) \subseteq I$ together with a derivative bound $|G'| \le L < 1$, the Banach fixed-point theorem gives both existence and uniqueness of the fixed point in $I$. The public interval contraction certificate [4] proves $G(I) \subseteq \mathrm{int}(I)$ and $L \le 0.0724$ for both readout modes by a direct Banach argument in mean-value (centered) form, with `mpmath.iv` outward rounding on every elementary operation and with the SU(2)/SU(3) edge-sum tails bounded by geometric majorants, so the enclosure covers the infinite-cutoff sums. The certified unique fixed points are $\alpha^{-1} = 136.994835177413\ldots$ for the source map (enclosure width $7.2 \times 10^{-24}$) and $\alpha^{-1} = 137.035660136946577\ldots$ for the gauge-width map. The certificate certifies the declared numerical map; it does not relate either fixed point to the measured constant. Both roots lie outside the measured comparison basin. The global at-most-one statement is discharged on the declared analytic domain ($\alpha^{-1} \in [100, 200]$): $\sup|g'| < 1$ on all 256 certified pieces, both readout maps, zero exceptional set (domain-global uniqueness certificate in the same artifact bundle), so each map has exactly one fixed point on the domain. A maximal-domain extension certificate closes the exterior. An envelope lemma built from the declared solver windows (the pixel scan window $\alpha_U \in [0.02, 0.08]$ and the $m_Z$ log-grid window $\ln(\mu_U/m_Z) \in [0, 50]$) together with the $m_Z$-closure identity certifies $1/\alpha_3(m_Z) \ge 4$ and a positive inverse-alpha readout floor on the whole analytic domain, and an interval sweep of the full pixel window certifies that every window-consistent fixed-point candidate of either readout map lies inside the declared solver interval (global uniqueness extension certificate). Each declared map therefore has exactly one fixed point on its maximal analytic domain, and the set of fixed points outside the declared interval is empty; the headline enclosures above are unchanged.

A source-only theorem requires a source-derived Ward-current spectral measure, the source Jacobi kernel, electromagnetic normalization, a same-scheme finite remainder, a target-independent rule selecting one physical map, a typed construction that identifies its two sides as the same quantity, a dependency graph excluding Thomson-target leakage, a full contraction interval for the selected map, and a prediction interval narrower than the adopted direct measurement. For the comparison standard used here, the direct rubidium recoil result of Morel, Yao, Cladé, and Guellati-Khélifa reports $\alpha^{-1} = 137.035999206(11)$ with 81 parts-per-trillion relative precision [22]. The operational receipt is therefore

$$\mathrm{rad}(\alpha_{\mathrm{pred}}^{-1}) < 1.1 \times 10^{-8}$$

in inverse-alpha units for the cited direct rubidium recoil interval.

## 26 An independent entangled-pair spectral candidate

Wissner-Gross gives an exact algebraic candidate for the Thomson coupling from a finite entangled-pair construction [31]. Its scalar can be checked independently of the proposed physical identification. For four slots of dimensions $(3, 2, 3, 2)$, traceless letters on unordered slot pairs give 169 product records. Three additional exchange-odd directions give a 172-dimensional nonzero feature space. The two commuting signed exchanges $X, Y$ satisfy $X^2 = Y^2 = I$. For

$$K = \frac{(2I - X)(3I - Y)}{2},$$

the joint projectors have gains $1, 2, 3, 6$ and ranks $79, 48, 42, 3$. The complete declared feature normalization is $(3^2 - 1)(2^2 - 1)(3 \cdot 2)^4 = 31104$. Thus

$$\alpha_{\mathrm{pair}} = \frac{79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}}{31104}, \qquad \alpha_{\mathrm{pair}}^{-1} = 137.035999189566155\ldots.$$

The executable reconstruction checks the involutions, integer projector identities, ranks and gains. It fits no continuous coefficient. These are properties of the stated finite operator and readout normalization.

The six-dimensional matrix algebra and a 24-dimensional mixed traceless space are compatible finite ingredients for comparison with OPH. Equal dimensions alone do not identify a state, a current representation or a photon two-point function. Matching invariance and reading the identity as one also do not uniquely select normalized trace: the normalized Bell projector state and the tracial state share those properties and evaluate the Bell projector differently. The normalized trace is therefore retained as an explicit readout choice in this comparison.

A complete OPH instrument retains the complementary Bell outcome. For two flat $3 \times 2$ pairs and the canonical qutrit Bell projection, its success probability is $1/6$, with failure probability $5/6$. Independently embedded flat supports need not contract to a flat pair: a tested pair of flat factors gives reduced eigenvalues $1/5, 4/5$ after contraction. Aligned supports or an adjusted contraction avoid that example, and are additional hypotheses. These controls bound the reuse of the pair construction without changing its canonical scalar arithmetic.

If the independent same-cell identification $P = \varphi + \sqrt{\pi}\,\alpha(0)$ is granted and $\alpha(0) = \alpha_{\mathrm{pair}}$, substitution gives

$$P_{\mathrm{pair}} = 1.630968209402773261\ldots.$$

This is a conditional coordinate, not a fixed-point theorem for either trial map analyzed in this paper. The two interval-certified trial-map roots retain their own hypotheses and endpoint tests. Substituting another coupling value does not modify those maps or supply their missing transport.

The same conditional coordinate can be checked across scales without a new fit. Exact rational square-root enclosures and alternating-series bounds for $\pi$ reproduce $\alpha_{\mathrm{pair}}^{-1}$ and propagate it through the declared relation $n_s = 1 - P/48$. Relative to the CODATA 2022 inverse-alpha reference, the scalar differs by approximately 0.5984 reported standard uncertainties. Its tilt is $0.966021495637442\ldots$, only $2.47 \times 10^{-14}$ above the CODATA-substituted tilt. Both ends of the rational enclosure give the same weighted lower-tail count $3114/508985$ in the retained Planck+ACT

DR6+lensing+DESI BAO posterior projection. This count is a posterior diagnostic, not a frequentist significance; the numerical enclosures bound arithmetic error, not physical model error. The comparison therefore leaves that conditional cosmological result unchanged. The independent calculation and the joint-construction scope check are part of the executable model's empirical diagnostics [12]. Neither a close coupling value nor a separately supplied oscillator or Weyl–Higgs action identifies one physical model across these measurements.

For a physical coupling claim, the same observer construction must identify the conserved electromagnetic current, its charge normalization, a massless photon pole, and the appropriate dressed-to-bare residue or equivalent low-momentum response, with controlled running and thresholds. The finite candidate supplies no such OPH attachment. Its numerical proximity to the experimental value is a retrospective comparison and is not a prospective prediction record. The calculation and its controls are self-contained in the executable model [12]; its elementary normalization and pixel-substitution algebra is formalized in the repository's Lean library.

### 26.1 A conditional combined gauge branch

The exact matter census also supplies representation indices $k = (10/3, 2, 2)$. If this column is identified with the physical kinetic ray, and the three-family, one-doublet Standard Model one-loop column is imported, write

$$(x_Y, x_2, x_3) = c(10/3, 2, 2) + t(41/6, -19/6, -7), \qquad x_i = \alpha_i^{-1}.$$

Adding the declared same-scale tree relation $x_Y + x_2 = \alpha_{\text{pair}}^{-1}$ and representation-entropy relation $\ell_2(\alpha_2) + \ell_3(\alpha_3) = P_{\text{pair}}/4$ gives an isolated positive branch. The entropy functions here use the stated representation dimensions and quadratic Casimirs with weights proportional to $d_R \exp(-4\pi^2 \alpha C_2(R))$. A monotonicity and tail certificate encloses the unique root in

$$5.002666247 < t < 5.002666250.$$

Its dimensionless couplings are approximately $\alpha_Y = 0.00922782$, $\alpha_2 = 0.03488203$, and $\alpha_3 = 0.10536109$. The electroweak quadratic mass form then has one photon null direction and charged-to-neutral ratio 0.79079908, per unspecified vacuum norm. These values consume the declared kinetic ray, one-loop census, entropy law and same-scale matching. They are not measured $Z$-scale predictions: Thomson-to-running transport, thresholds, physical kinetic normalization, vacuum scale and pole matching are not supplied. The executable coefficient and mass-form calculations retain those inputs explicitly.

## 27 Reproducibility

The executable fixed-point calculation is available in the public repository [4]. From the code directory cited there, the main human-facing CLI is

```
python3 derive_p.py --color always
```

To print only the pure source value:

```
python3 derive_p.py --no-hadron-closure
```

To emit machine-readable output:

```
python3 derive_p.py --json --output runtime/report.json
```

To run the raw PDG/CERN diagnostic discussed in Section 22:

```
python3 fine_structure_fixed_point_demo.py --compare-alpha-inv 137.035999177
```

Informally: the command line separates the pure source value, the CODATA/NIST comparison value, and the diagnostic values. The mathematical paper uses the same separation.

## 28 Complete Source-to-Fixed-Point Chain

| Step | Mathematical object | Scope |
|---|---|---|
| Overlap rule | Eq. (1) | Agreement condition; realized by native repair in the executable specification. |
| Pixel variable | $P = a_{\mathrm{cell}}/\ell_\star^2$ | Defined. |
| Golden balance | $\varphi = (1+\sqrt{5})/2$ | Defined. |
| Boundary width | $\alpha_{\mathrm{ext}} = (P-\varphi)/\sqrt{\pi}$ | Defined. |
| Inside readout | $\alpha_{\mathrm{in}} = 1/A_{\mathrm{Th}}(P)$ | Defined. |
| Closure | $P = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ | Defined. |
| Source scale | $M_U = E_P \mathrm{e}^{-2\pi} P^{1/6}$ | Defined and calculated. |
| Cell scale | $E_{\mathrm{cell}} = E_P/\sqrt{P}$ | Defined and calculated. |
| Transmutation | $v = E_{\mathrm{cell}} \exp[-2\pi/(\beta_{\mathrm{EW}}\alpha_U)]$ | Defined and calculated. |
| Gauge running | Eq. (17) | Defined and calculated; the coefficient triple is the one-loop MSSM set, a counted structural selection. |
| $Z$-scale self-consistency | Eq. (20) | Defined and calculated. |
| Heat-kernel closure | Eq. (27) | Defined and calculated. |
| Electroweak anchor | $A_Z = \alpha_{\mathrm{em}}^{-1}(m_Z^2; P)$ | Defined and calculated. |
| Charged spectrum | Eqs. (36) and (39) | Declared phenomenological transport input; not an OPH-derived mass prediction. |
| Fermion kernel | Eqs. (40) and (42) | Exact one-loop kernel. |

| Step | Mathematical object | Scope |
|---|---|---|
| Screening ansatz | Eq. (46) | Declared selection of the quark continuation; asserted rather than derived, with no perturbative derivation. |
| Calculated endpoint | $A_{\mathrm{calc}} = A_Z + \Delta_{\mathrm{calc}}$ | Raw no-hadron diagnostic conditional on the supplied charged-spectrum continuation. |
| Bare-$\alpha_U$ closure | $A^{\mathrm{fp}}_{\alpha_U}$, Eq. (63) | Mixed-provenance display packet; the certified gauge-width fixed point is $137.035660136946577\ldots$, without constituting a derivation of $\alpha$. |
| Required hadronic correction | $\Delta^{\mathrm{fp}}_{\mathrm{H,req}}$, Eq. (66) | CODATA/NIST comparison residual attributed to the missing same-scheme hadronic endpoint transport. |
| Hadronic/scheme residual | $R_Q = A_{\mathrm{Th}} - A_{\mathrm{calc}}$ | Isolated endpoint contribution. |
| CODATA/NIST endpoint | $A_{\mathrm{Th}} = 137.035999177(21)$ | Measured endpoint comparison value. |
| Two-$P$ provenance | $P_{\mathrm{root}}$ versus $P_{\mathrm{C}}$ | Source-branch root and measured-endpoint comparison pixel are distinct; $P_{\mathrm{C}}$ inherits CODATA/NIST by definition. |
| Raw PDG diagnostic | Eq. (79) | Comparison diagnostic in a different endpoint convention. |
| Interval theorem | Banach/contraction certificate for full $G$ | Discharged for the declared numerical map (public certificate, both readout modes); required for the full source map with the hadronic term. |

## 29 Conclusion

Conditional on target-independent map selection, the declared same-cell bridge, and same-scheme endpoint transport, the fine-structure constant is the coupling that makes one OPH screen cell internally consistent. The outside reading says that the cell is displaced from the golden-ratio balance by $(P-\varphi)/\sqrt{\pi}$. The inside reading sends the same $P$ through the declared source map, gauge closure, electroweak anchor, and Ward-projected electromagnetic transport, producing $1/A_{\mathrm{Th}}(P)$. Neither the exact fixed-point mathematics nor the source certificates construct that bridge or select either certified map. The construction fits no continuous parameter, and each declared incomplete map has a machine-certified unique fixed point on its declared domain. The map selector, same-cell construction, hadronic source payload, and detached no-target-leak certificate are premises this construction does not supply, so the residuals are diagnostics rather than a derivation of $\alpha$.

The pure source calculation computes the root witness

$$\alpha_{\mathrm{root}}^{-1} = \frac{136.994835177412937295289429464436902857561206151035393184502}{},$$

the interval-certified unique fixed point of the declared source map (enclosure width $7.2 \times 10^{-24}$, shipped interval certificate). The CODATA/NIST measured endpoint value is

$$\alpha^{-1}(0) = 137.035999177(21),$$

with

$$P = 1.630968209403959324879279847782648941\ldots.$$

Combining the source root with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel gives the mixed-provenance no-hadron diagnostic

$$A_{\alpha_U}^{\mathrm{fp}} = 137.035959513608567790289429464437\ldots,$$

which is not a fixed point of a single declared map. The certified self-consistent gauge-width fixed point is $\alpha^{-1} = 137.035660136946577\ldots$, $2.5 \times 10^{-6}$ below the measured value, about $1.6 \times 10^{4}$ measurement sigma. The residual gap is CODATA-minus-diagnostic bookkeeping, not an emitted low-energy hadronic calculation. The physical pure-source theorem requires the Ward-projected hadronic spectral measure, same-scheme transport, target-independent map selection, the typed same-quantity bridge, and the interval-certificate step. The endpoint audit holds $P$ and the declared branch fixed. It rejects only the reading of the source root as a complete endpoint with zero missing correction; it assigns no theory sigma, performs no fit and does not recalibrate $P$ [6]. The public-data audit withholds a new prediction claim for the fine-structure constant [7].

## Declarations

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**Data Availability declaration.** No experimental data were generated. The manuscript cites the 2022 CODATA/NIST inverse fine-structure constant [26, 27], public hadronic cross-section and vacuum-polarization references [24, 20, 21], and the OPH source materials cited below. The TeX source, OPH paper sources, and fixed-point calculation code are available in the public repository [4].

**Code Availability declaration.** The fixed-point code and runtime artifacts are available in the public repository [4].

**Ethics declaration.** Not applicable.

**Competing Interests declaration.** The authors declare no competing interests.

## Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Primary mathematical and physical sources [31, 30].
- Running-alpha and hadronic vacuum-polarization literature [21, 29].

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